ProteinQ - QML Model to Predict SIRT6 Protein Bindings

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Track: Quantum Machine Learning, Problem 3: Build and Benchmark a Variational Quantum Classifier (VQC) ·

Problem and Motivation

Drug discovery is one of the fields named most often as an early use for quantum computing. This project tests whether a small quantum model can learn a real chemistry task today, and how it compares to standard classical models. SIRT6 is a protein studied as a drug target, where its expression can lead to oncogenesis [2]. SIRT6 allows cancerous cells to evade apoptosis (cell death) by deacetylating histone H3K9 [2]. Overall, the activation of SIRT6 in specific situations could promote conditions favorable for cancerous cells, prompting our team to develop a quantum algorithm to screen small molecules for optimal inhibitors. Screening candidate molecules for strong binding is slow and costly, so a fast model that ranks candidates first saves lab time. Binding free energy (BFE) measures how tightly a molecule binds: low BFE means a strong binder and a better drug candidate. We used the UCI SIRTUIN6 dataset: 100 molecules, 6 descriptors (SC-5, SP-6, SHBd, minHaaCH, maxwHBa, FMF), labeled High_BFE or Low_BFE [1].

Success would be a team-built VQC that beats the 52% majority-class guess on held-out data, comes within about 0.05 macro F1 of the best classical baseline of similar size, and gives similar results across random seeds.

Method

Setup: The SIRT6 dataset has 100 molecules, 6 features and 2 classes (High_BFE / Low_BFE). We used a stratified split of 75 train / 25 test, with an inner 60 / 15 train / validation split for tuning. Inputs were standardized and encoded as angles via $2 \cdot \arctan(x)$. The VQC uses RY encoding, trainable RY layers with a CNOT chain, one measured qubit, and COBYLA with 100 iterations.

Research Question (RQ) 1 : How do training settings affect VQC performance?

Optimizer Step Size : To see how step size in the optimizer affects training, we trained a 2-layer VQC with three starting step sizes (that being rhoberg = 0.1, 0.5, and 1.0) keeping the random seed at 42. The smallest step performed the worst, spitting out an accuracy of 0.60 and a training loss of 0.557. Both 0.5 and 1.0 reached 0.80 accuracy with training loss being 0.431. Given that

these last two performed equally, 0.50 was used based on validation macro F1 and used it in the following experiments.

Circuit depth : The step size of 0.5 and seed 42 was then used to test how the number of variational layers affects performance. The 2 layer circuit was concluded to have the highest validation accuracy of 0.8. Both 1 and 3 layer circuits achieved an accuracy of 0.60. So 2 layer was selected.

Random Initialization : Lastly we checked whether the random starting point affects the results by training our 2 layer, rhoberg = 0.5 model with three seeds of 7, 21, and 42. The validation accuracy was 0.667, 0.800 and 0.800 respectively. The starting point alone changed accuracy by more than 13 percentage points. Across these 3 experiments, the highest validation macro F1 of 0.7847 came from a 2 layer, rhoberg = 0.5 and seed 42 model. This was used as our final VQC

RQ 2 : How does the selected VQC compare with classical models on all 6 features?

VQC and SVM final test : To evaluate the final models, the selected VQC was retrained on all 75 training molecules and tested it one time on the 25 holdout molecules. We performed the same for a RBF-SVM as a baseline to use. The VQC reached a test accuracy of 0.72 and a macro F1 of 0.72. All the while SVM reached 0.84 accuracy and 0.838 macro F1. The SVM outperformed the VQC.

Logistic Regression Baseline : Logistic regression was trained on the same split as a second baseline. It reached a test accuracy of 0.84 and a macro F1 of 0.84, matching the SVM. This model also performed better than the VQC.

Three Model Comparison : Comparing the three models, VQC reached a test accuracy of 0.72 while logistic regression and SVM both reached 0.84. This 0.12 gap means the classical models correctly classified 3 more of the 25 test molecules. So when all 6 features were used, the VQC did not outperform classical methods.

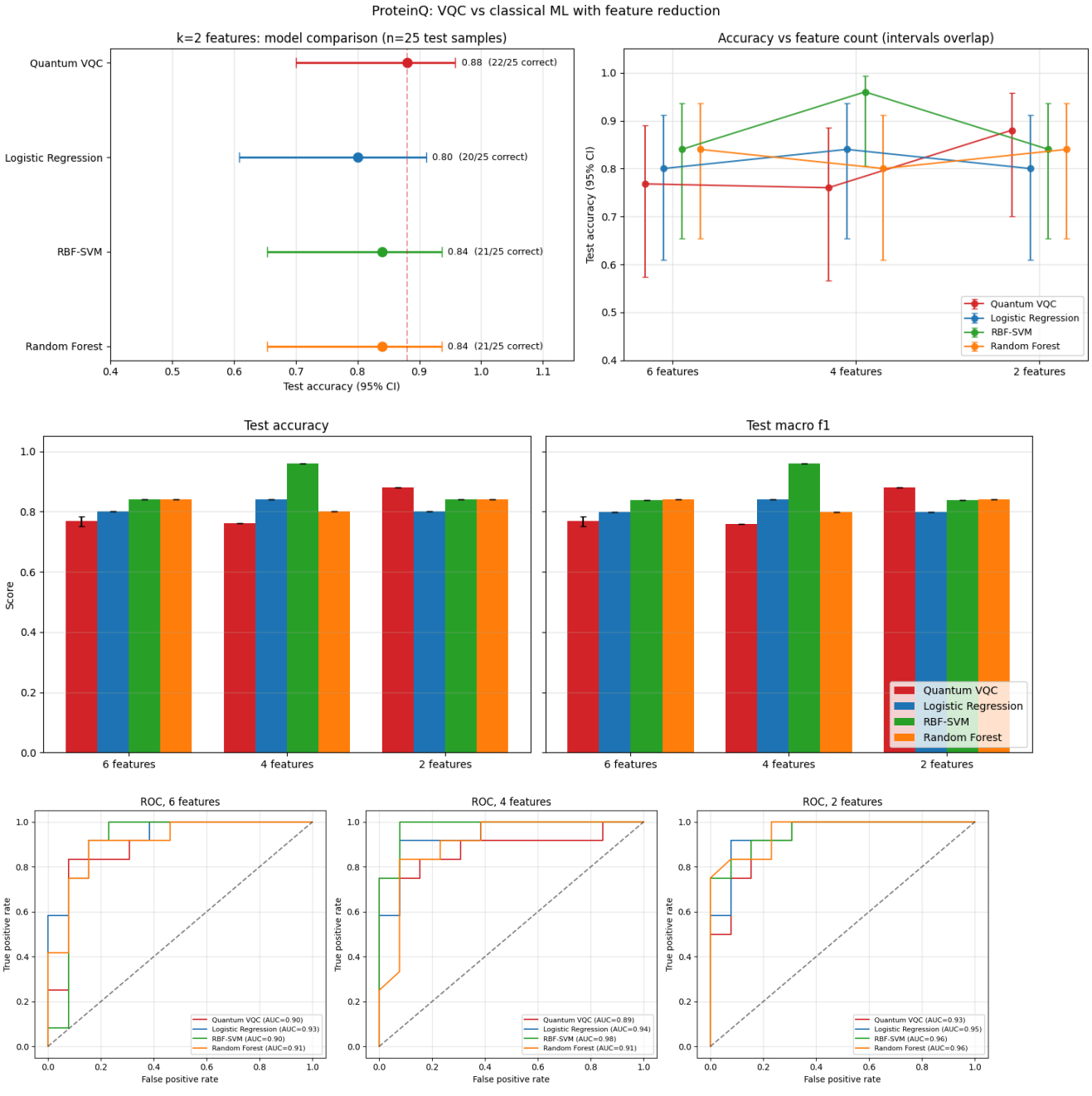
Question 3 : How does feature reduction affect the VQC and classical models?

Performance with 6, 4, and 2 features: To test the effect of feature reduction, we ranked the features by random forest importance (SP-6 > FMF > miniHaaCH > SC-5 > maxwHBa > SHBd) and trained each model on the top 6, 4, and 2 features. As features were removed, VQC improved from 0.768 to 0.880 when it went from 6 to 2 features respectively. The classical models stayed between 0.80 and 0.84, with SVM reaching 0.96 at one point with 4 features.

Training behavior and train vs test accuracy We also compared training and test accuracy for one VQC at each feature count. At 6 and 4 features the VQC scored 0.813 on training data and 0.760 on test data. With 2 features, it then scored higher on test at 0.880 and on training 0,840. The SVM's training accuracy fell as features were removed.

How certain are the results? : Because the test set had only 25 molecules, we calculated 95% wilson confidence intervals for each model at 2 features. The VQC's interval was 0.700 - 0.958, compared with 0.609 - 0.911 for logistic regression. Then finally 0.653 - 0.936 for both the SVM and random forest. Since all intervals overlap, the VQC's higher score at 2 features is not strong evidence that it outperforms classical models.

Results



For details on graphs, please refer to the notebook in the GitHub repository section.

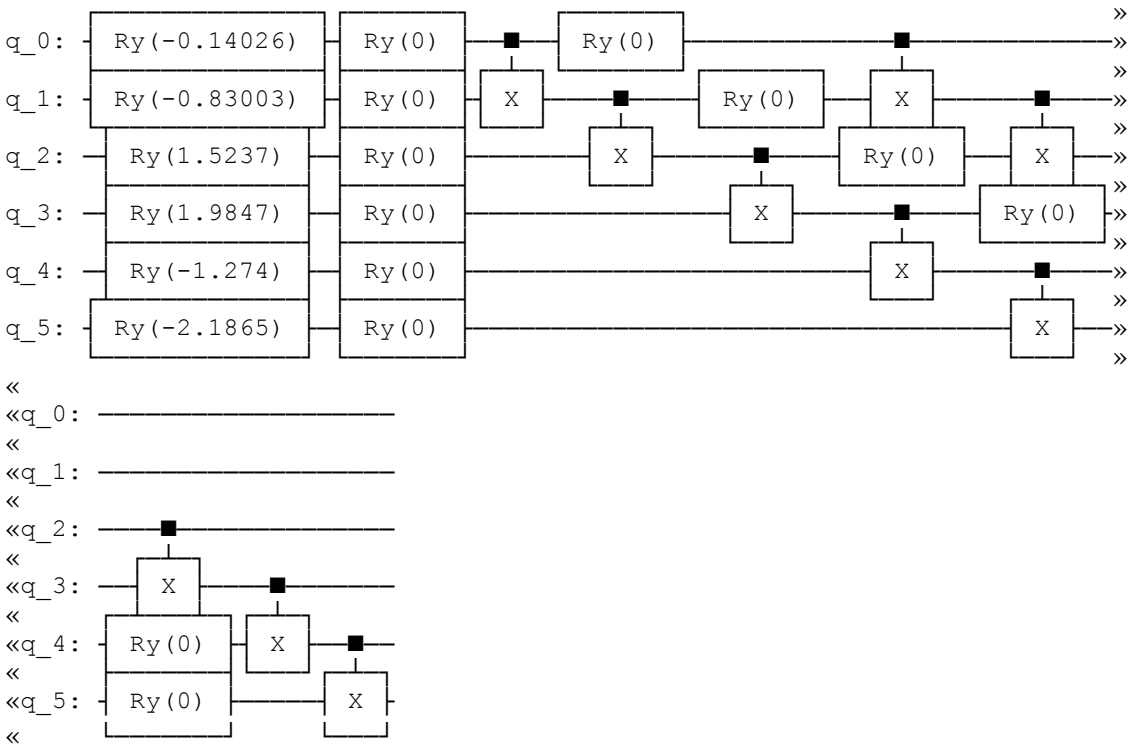
Hardware

We used the IBM platform and the full results will be inside our GitHub repository.

Honest Limits

No quantum advantage is claimed. Classical models beat the VQC at 6 and 4 features and tie it at 2, where with 25 test molecules (4 points each) it is neither shown better nor worse. They also train in milliseconds with no random start to tune.

Circuit Diagram:



References

[1] SIRTUIN6 small molecules dataset, UC Irvine Machine Learning Repository.

[2] Fiorentino, F., Mai, A., & Rotili, D. (2021). Emerging therapeutic potential of SIRT6 modulators. Journal of Medicinal Chemistry, 64(14), 9732–9758. https://doi.org/10.1021/acs.jmedchem.1c00601

Repository: Full notebook with all training curves and confusion matrices, exported CSV results, and the project plan.

https://github.com/hareem-arshad/quantum-ml-protein-analysis

AI Usage Report:

We only used ChatGPT/Claude's chat features to clarify concepts, improve documentation, and make suggestions on the code. Our team performed a human review on the AI-generated answers subsequent to every suggestion.

AI's mistake: Some AI suggestions were general and not specific to our experimental setup, so they could not be verified for our project. We caught these mistakes after reviewing the suggestions and testing and implementing them.